3.2 Nonlinear dynamics propagation of B747 based on RK4

As can be seen from Section 3.1, the planar flight dynamics of the Boeing 747 exhibit nonlinear coupling: the position equations are linked to the heading angle through trigonometric functions, while the heading angle itself is driven by the bank angle. In addition, the lift coefficient involved in the drag model D is also nonlinearly coupled, making it impossible to obtain a closed-form analytical solution for this set of equations. Therefore, a high-accuracy numerical integration method is required to propagate the nonlinear states.

3.2.1 The fourth-order Runge-Kutta method

RK4 method constructs a high accuracy approximation at each time step $\left[t_k, t_{k+1}\right]$ by sampling the state derivatives at four points within the integration interval. Let the state equation is $\dot{x} = f\left(x, u\right)$ and the control input remains constant over each time step under a zero-order hold. The update equations of RK4 are:

$$k_1 = f\left(x_k, u_k\right)$$

$$k_2 = f\left(x_k + \frac{\Delta t}{2} k_1, u_k\right)$$

$$k_3 = f\left(x_k + \frac{\Delta t}{2} k_2, u_k\right)$$

$$k_4 = f\left(x_k + \Delta t k_3, u_k\right)$$

$$x_{k+1} = x_k + \frac{\Delta t}{6}\left(k_1 + 2k_2 + 3k_3 + k_4\right)$$

The coefficients are chosen so that the RK4 update matches the Taylor expansion of the exact solution up to fourth order:

$$x\left(t_{k+1}\right) = x\left(t_k\right) + \Delta t f\left(x_k, u_k\right) + \frac{\left(\Delta t\right)^2}{2!} f'(x_k, u_k) + \frac{\left(\Delta t\right)^3}{3!} f^{(2)}(x_k, u_k) + \frac{\left(\Delta t\right)^4}{4!} f^{(3)}(x_k, u_k) + O\left(\left(\Delta t\right)^5\right)$$

For comparison, the forward Euler method $x_{k+1} = x_k + \Delta t f\left(x_k, u_k\right)$ retains only the first-order term, with local error of $O\left(\left(\Delta t\right)^2\right)$ and global error of $O\left(\Delta t\right)$. In terms of global

error, halving the step size, the error of RK4 is reduced to $\frac{1}{16}$ of its original value, whereas the error of the Euler method is reduced only to $\frac{1}{2}$.

3.2.2 Application of RK4 to the Five-Dimensional B747 State

The current state is $x_k = \left[x_k, y_k, \psi_k, \phi_k, v_k\right]^T$, and the control input is $u_k = \left[\phi_{cmd,k}, T_k\right]^T$

1) First slope k_1: instantaneous motion at the current time step

$$k_1 = \begin{bmatrix} v_k \cos\psi_k \\ v_k \sin\psi_k \\ g \tan\frac{\phi_k}{v_k} \\ \frac{\phi_{cmd,k} - \phi_k}{\tau_\phi} \\ \frac{T_k - D\left(\phi_k, v_k\right)}{m} \end{bmatrix}$$

k_1 is determined by the current state. In this equation, $k_{1.1}$ and $k_{1.2}$ represent the ground speed components the aircraft along the current heading angle ψ_k. $k_{1.3}$ is the turning rate generated by the current bank angle. For symmetric circular fight, with $\phi \approx 20.1^\circ, v = 210 m/s$, the value is approximately $0.0171 rad/s$ $k_{1.4}$ reflects the first-order respond of the bank angle towards the commanded value, and $k_{1.5}$ represents the acceleration generated by the difference between trust and drag.

2) Second slope k_2: corrected slope after half-step prediction

Using k_1, the state is advanced by half a step to $t_k + \frac{\Delta t}{2}$:

$$\tilde{x}^{(2)} = x_k + \frac{\Delta t}{2} k_1$$

For the B747 with $\Delta t = 0.05s$ and $\dot{\psi}_k = 0.0171 rad/s$:

$$\tilde{\psi}^{(2)} = \psi_k + 0.025 \times 0.0171 \approx \psi_k + 0.025^\circ$$

$$\tilde{\phi}^{(2)} = \phi_k + \frac{0.025 \times \left(\phi_{cmd} - \phi_k\right)}{\tau_\phi}$$

The small change in heading angle causes corrections in $\cos\tilde{\psi}^{(2)}$ and $\sin\tilde{\psi}^{(2)}$. k_2 captures the first-order correction effect of heading variation on the integration of position.

3) Third slope k_3: improved midpoint slope

$$\tilde{x}^{(3)} = x_k + \frac{\Delta t}{2} k_2$$

k_2 has been considered the change over the half step, so k_3 uses the prediction based on k_2 the resulting position more accurately.

4) Fourth slope k_4: endpoint slope

$$\tilde{x}^{(4)} = x_k + \Delta t k_3$$

k_4 evaluates the slope at the endpoint state predicted by k_3. In the final weighted average, k_1 and k_4 each account for $\frac{1}{6}$ of the total weight, while the k_2 and k_3 each account for $\frac{2}{6}$ of the total weight. Therefore, the midpoint contributes most to the integration accuracy. Compared with Euler method, RK4 method uses slope at different points to evaluate the influence on the position by changing heading, rather than only evaluating the slope at start point. The comparison results of RK4 and the Euler method are shown in Figure 3.1.

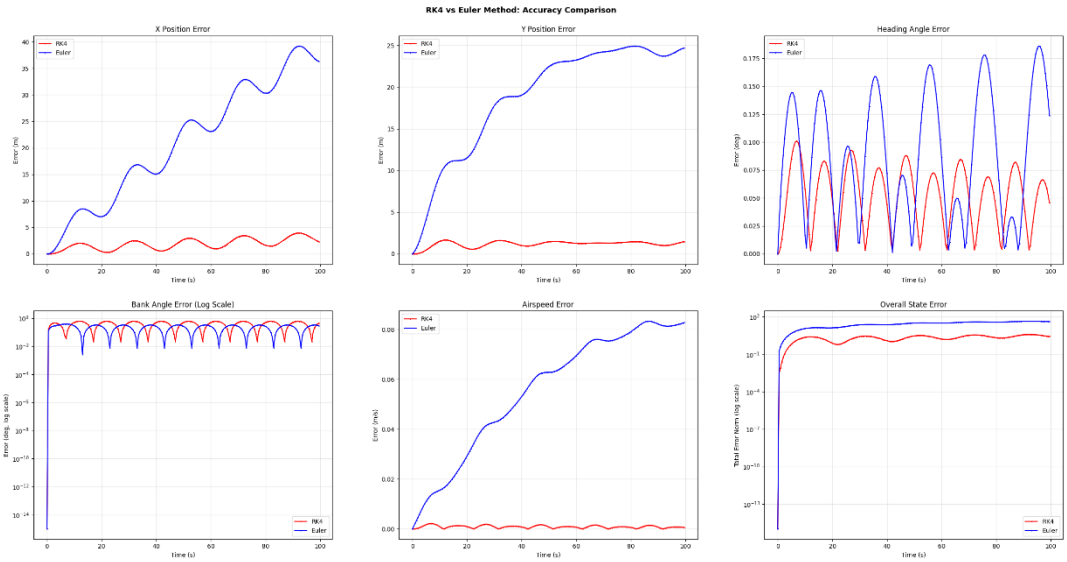

Figure 3.1 RK4 method vs Euler method

Figure 3.1 shows a comparison of the accuracy of the RK4 method and the Euler method at the same time length. The parameter settings in the simulation are shown in Table 3.1:

Table 3.1 Parameter settings for the RK4 method in simulation

parameter	value	physical meaning
Δt_{ref}	0.05s	generating truth values in RK4
Δt_{test}	0.5s	comparison using the same step size
T_{sim}	100s	total simulation time
ϕ_{cmd}	$15^\circ + 10^\circ \sin\left(\frac{2\pi t}{20}\right)$	time-varying control input

Results Analysis(这部分看要不要挪到其他章节，我先都写到一起，确定章节框架之后挪)

The simulation results demonstrate significant accuracy differences between the RK4 and Euler methods. In terms of position errors, the RK4 method maintains errors of approximately 3m in the x direction and 2m in the y direction, while the Euler method exhibits continuous error accumulation, reaching approximately 40m and 25m respectively by the end of the 100-second simulation. This yields an error ratio of approximately 13:1, a substantial difference attributable to the cumulative effect of heading angle integration errors on position states.

The heading angle error exhibits a distinct periodic oscillation pattern with a 20s period,

corresponding to the control input frequency. The RK4 method maintains an error amplitude of approximately $0.05-0.10^{\circ}$, while the Euler method exhibits an amplitude of approximately $0.10-0.20^{\circ}$, indicating that the Euler method's tracking precision for time-varying targets is approximately half that of RK4.

The bank angle error is presented on a logarithmic scale because the bank angle dynamics is a linear first-order system with relatively small error magnitudes. The RK4 error ranges from 10^{-2} to 10^{0} degrees, while the Euler error ranges from 10^{-1} to 10^{0} degrees, yielding an error ratio of approximately 10:1. The logarithmic scale enables clear visualization of both curves. Regarding airspeed error, the RK4 error is negligible, while the Euler error grows to approximately 0.08 m/s. This demonstrates that the velocity dynamics are relatively less affected by integration errors due to the equation's relative independence and the limited nonlinearity of the drag

model.

To compare the overall performance of the two methods, we plotted an overall state error figure. It is obvious that the RK4 method maintains a total error stable at approximately 3.6, while the Euler method's total error grows to approximately 53, resulting in a final error ratio of approximately 15:1. These results align perfectly with theoretical analysis: the RK4 method's global truncation error is $O\left((\Delta t)^{4}\right)$, while the Euler method's is $O(\Delta t)$. With a step size ratio of 1:10, the RK4 error relative to the reference solution should theoretically be approximately $\frac{1}{10}$ that of Euler, and the observed 10-15 fold difference in the simulation validates this theoretical prediction.

In conclusion, this comparative experiment thoroughly demonstrates the significant accuracy advantage of the RK4 method in handling the nonlinear B747 dynamics equations.

3.2.4 Step size selection

Choosing the step size requires consideration of two factors: numerical accuracy and numerical stability, which are related to the system dynamics of B747. As can be seen from Section 3.1, there are several different time scales. The bank angle response equation is:

$$\dot{\phi} = \frac{\phi_{\text{cmd}} - \phi}{\tau_{\phi}}$$

It is a standard first-order lag system, τ_{ϕ} represents the respond speed. When a step bank angle command is applied, the bank angle reaches approximately 63.2% of its total response after 2 seconds. Therefore, the first-order bank angle response is the fastest mode in the system. It determines the step size.

3.4 Gaussian Bayesian/MAP Estimation for B747 Flight Condition

In the actual flight, the controller can't directly access the exact state of the aircraft. The measurements provided by navigation sensors always contain noise. Therefore, this report intends to infer the real aircraft state and quantify the uncertainty.

3.4.1 Measurement model and noise parameters

The measurement model contains two parts: measurement from sensors and noise:

$$z_k = x_k + v_k$$

We assume that the noise follows a normal distribution: $v_k \sim N(0, R)$. As can be seen from 2.2, the continuous state is $x_k = [x_k, y_k, \psi_k, \phi_k, v_k]^T$. Let the measurement deviation: $\sigma_z = (40, 40, 1, 1, 2)$, and the physical meanings are shown in Table 3.2:

Table 3.2 physical meanings of each measurement deviation

state	deviation	physical meaning
x, y	40m	horizontal accuracy
ψ	1°	heading noise
ϕ	1°	bank angle noise
v	2m/s	uncertainty of airspeed tube

$\sigma_{\psi} = 1^\circ = \frac{\pi}{180} rad$, $\sigma_{\phi} = 1^\circ = \frac{\pi}{180} rad$. Therefore, the measurement noise covariance matrix is:

$$R=\begin{bmatrix} 40^2 & 0 & 0 & 0 & 0 \\ 0 & 40^2 & 0 & 0 & 0 \\ 0 & 0 & \left(\frac{\pi}{180}\right)^2 & 0 & 0 \\ 0 & 0 & 0 & \left(\frac{\pi}{180}\right)^2 & 0 \\ 0 & 0 & 0 & 0 & 2^2 \end{bmatrix}$$

Let the process noise deviation: $\sigma_q=(2,2,0.2,0.2,0.5)$, and the physical meanings are shown in Table 3.3.

Table 3.3 physical meanings of each process noise deviation

state	deviation	physical meaning
x, y	2m	wind disturbance on position
ψ	0.2°	heading disturbance caused by wind direction variation
ϕ	0.2°	bank angle disturbance caused by turbulence
v	0.5m/s	wind speed variation on airspeed

Therefore, the measurement noise covariance matrix is:

$$\mathrm{Q}=\begin{bmatrix} 2^2 & 0 & 0 & 0 & 0 \\ 0 & 2^2 & 0 & 0 & 0 \\ 0 & 0 & \left(0.2\times\frac{\pi}{180}\right)^2 & 0 & 0 \\ 0 & 0 & 0 & \left(0.2\times\frac{\pi}{180}\right)^2 & 0 \\ 0 & 0 & 0 & 0 & 0.5^2 \end{bmatrix}$$

3.4.2 Bayesian update

1) Prior:

At the step k, based on the estimate from the previous step and dynamic model, the

prior distribution of the position is:

$$x_k^- \sim \mathrm{N}\left(\hat{x}_k^-, P_{xx,k}^-\right)$$

Where $\hat{x}_k^-$ is the predicted position, and $P_{xx,k}^-$ is the uncertain of the prediction.

2) Likelihood:

The position measurement provided by the sensor is:

$$z_k = x_k + v_k, \quad v_x \sim \mathrm{N}\left(0, R_{xx}\right)$$

Therefore, the likelihood function is:

$$p\left(z_x \mid x\right) \propto e^{\left(-\frac{\left(z_x - x\right)^2}{2R_{xx}}\right)}$$

Posterior: By Bayes' theorem, $p\left(x \mid z_x\right) \propto p\left(z_x \mid x\right) p\left(x\right)$. The prior, the likelihood and posterior are Gaussian. Merge the quadratic terms in x in the exponent:

$$-\frac{1}{2}\left[\frac{\left(x - \hat{x}_k^-\right)^2}{P_{xx,k}^-} + \frac{\left(z_x - x\right)^2}{R_{xx}}\right]$$

Expanding into a perfect square form:

$$\frac{1}{P_{xx,k}^+} = \frac{1}{P_{xx,k}^-} + \frac{1}{R_{xx}}$$

The posterior mean is the precision-weighted average:

$$\hat{x}_k^+ = P_{xx,k}^+\left(\frac{\hat{x}_k^-}{P_{xx,k}^-} + \frac{z_x}{R_{xx}}\right)$$

Extending this result to the five dimensional state vector, under the assumption that each dimension is independent, $\mathrm{H} = \mathrm{I}$, and both R and P^- are diagonal matrices.

$$\mathrm{K}_k = P_k^-\left(P_k^- + R\right)^{-1}$$

$$x_k^+ = x_k^- + K_k\left(z_k - x_k^-\right)$$

$$P_k^+ = \left(I - K_k\right) P_k^-$$

3.4.3 MAP estimation

Define the MAP objective function as:

$$L_{MAP} = \frac{1}{2}\left(x_k^+ - x_k^-\right)^T \left(P_k^-\right)^{-1} \left(x_k^+ - x_k^-\right) + \frac{1}{2}\left(z_k - x_k^+\right)^T R^{-1} \left(z_k - x_k^+\right)$$

In this equation, the first term is the model deviation coast: the farther the estimate deviates from the dynamic prediction, the larger the cost. The weight is determined by the $\left(P_k^-\right)^{-1}$.

The second term is the measurement fitting cost: the farther the estimate deviates from the sensor value, the larger the cost. The weight is determined by the R^{-1}.

For the B747 case, let the prior variance is $P_{xx}^- = 50m^2$, and the measurement variance is $R_{xx} = 1600m^2$. At that moment, the model confidence is larger than measurement confidence. The MAP optimization result will be more inclined toward the model prediction.

Letting the gradient of L to zero:

$$\nabla_x L = \left(P_k^-\right)^{-1} \left(x - x_k^-\right) + R^{-1}\left(x - z_k\right) = 0$$

Solving this equation:

$$x_{MAP} = \left[(P_k^-)^{-1} + R^{-1}\right]^{-1} \left[\left(P_k^-\right)^{-1} x_k^- + R^{-1} z_k\right] = x_k^+$$

The Hessian matrix is: $\nabla^2 \mathrm{L} = (P_k^-)^{-1} + R^{-1}$. Since both P_k^- and R are covariance matrices: $P_k^- \succ 0 \rightarrow \left(P_k^-\right)^{-1} \succ 0$ and $R \succ 0 \rightarrow R^{-1} \succ 0$. Therefore, the sum of positive definite matrices is still positive definite: $\nabla^2 \mathrm{L} \succ 0$. The x_{MAP} is the unique global minimum.

3.4.5 The implementation of prediction step

State prediction uses the RK4 integrator to propagate the posterior estimate from the precious step forward through the nonlinear dynamics:

$$x_k^- = f_{RK4}\left(x_{k-1}^+, u_{k-1}\right)$$

The complete covariance propagation is:

$$P_k^- = F_k P_{k-1}^+ F_k^T + Q, \quad F_k = \frac{\partial f_{discrete}}{\partial x}\Big|_{x_{k-1}^+}$$

F_k is the Jacobian matrix of the discrete-time dynamics. Using the simplified version $F \approx I$, when Δt is sufficiently small. The simplified form of the covariance prediction is : $P_k^- = P_{k-1}^+ + Q$.